

vGPU

The vGPU module is module ID 2. On a supported NVIDIA vGPU host, it exposes inventory and telemetry for the physical GPUs that can host vGPUs, the vGPU types available on those GPUs, and active vGPU instances.

When vGPU monitoring is useful

This module is intended for host-side visibility in a virtualized GPU environment. It is useful when an administrator needs to correlate a virtual machine with its vGPU, confirm the guest driver or license state, measure the framebuffer assigned to an instance, or investigate encoder and framebuffer-capture activity.

DCGM complements the virtualization platform in this workflow:

1. The hypervisor or vGPU management stack creates and assigns a vGPU.
2. That management inventory supplies the active vGPU instance ID and its owning VM.
3. DCGM watches the physical GPU and vGPU-instance fields while the VM runs.
4. An operator correlates DCGM telemetry with the VM, guest, and physical GPU when investigating capacity, licensing, or performance.

DCGM does not create VMs, provision vGPU instances, or replace the vGPU management interfaces. The information is available on the host; a DCGM instance inside a guest sees the GPU interface presented to that guest rather than the host’s complete vGPU inventory.

Entities and fields

An active vGPU instance is a `DCGM_FE_VGPU` entity and uses the command-line selector `vgpu:<id>`. Useful instance fields include:

Common vGPU instance fields

Field ID	Field name	What it answers
520	<code>DCGM_FI_DEV_VGPU_VM_ID</code>	Which VM identifier owns the vGPU?
521	<code>DCGM_FI_DEV_VGPU_VM_NAME</code>	What VM name is reported by the virtualization stack?
522	<code>DCGM_FI_DEV_VGPU_TYPE</code>	Which vGPU type ID is assigned?
523	<code>DCGM_FI_DEV_VGPU_UUID</code>	What UUID identifies the vGPU instance?

Field ID	Field name	What it answers
524	DCGM_FI_DEV_VGPU_DRIVER_VERSION	Which guest vGPU driver version is reported?
525	DCGM_FI_DEV_VGPU_MEMORY_USAGE	How much framebuffer is the instance using?
526	DCGM_FI_DEV_VGPU_LICENSE_STATUS	Is the instance licensed (1) or unlicensed (0)?
527	DCGM_FI_DEV_VGPU_FRAME_RATE_LIMIT	What frame-rate limit applies to the instance?

Physical-GPU vGPU inventory uses GPU-scoped fields 500 through 513, including virtualization mode, active instance IDs, and supported or creatable vGPU types. Fields 520 through 534 are scoped to an individual vGPU instance. Always confirm the definitions and scope in the field reference for the installed release.

Example: inspect an active vGPU

DCGM does not currently provide a dedicated `dcmgi` command that enumerates vGPU entity IDs. Obtain the active ID from the hypervisor or vGPU management inventory. Suppose that inventory reports instance ID 17. Stream identifying and operational fields for that instance as follows:

```
$ dcmgi dmon --entity-id vgpu:17 \
  --field-id 521,524,525,526,527 --count 5
```

This sample can establish whether the expected VM is attached, whether the guest driver and license are visible, and whether framebuffer consumption is changing. A license status of 0 is an actionable vGPU licensing signal; N/A is different and can mean that the field is unsupported, unavailable, not yet sampled, or paired with the wrong entity.

For several instances, either list them directly or create a reusable group:

```
$ dcmgi group --create tenant-a --add vgpu:17,vgpu:23
```

```
$ dcmgi dmon --group-id <group-id> --field-id 525,526 --count 10
```

Replace `<group-id>` with the ID printed by the create command. Groups are selectors in the running host engine; they do not establish VM isolation or vGPU ownership.

Limits of the command-line view

Some vGPU data, including per-process utilization and encoder or framebuffer- capture session structures, is represented by binary fields. `dcmi dmon` does not render binary field payloads and displays them as `N/A`. Use the C/C++ field APIs or the vGPU management interfaces when the workflow needs those structured records.

Field availability also depends on the NVIDIA vGPU software release, physical GPU, active instance, and privileges. A numeric field ID existing in the DCGM catalogue does not guarantee that every vGPU environment can supply it.

Further reading

- [dcmi dmon](#) documents monitoring syntax, field output, and unavailable values.
- [dcmi group](#) documents reusable entity groups and the `vgpu` selector.
- [dcmi discovery](#) documents what `dcmi` can and cannot enumerate.



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